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MACHINE LEARNING CHALLENGES IN THE CALIBRATION OF THE CPB'06 YIELD CRITERION

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Summary. *Advanced anisotropic yield criteria such as the Cazacu-Plunkett-Barlat (CPB'06) model pose significant challenges for parameter identification due to their nonlinear structure and high dimensionality. This work investigates the machine learning-based identification of CPB ´06 parameters from numerically generated biaxial tensile tests on cruciform specimens. XGBoost and feedforward neural networks (FNNs) are compared using a physically constrained dataset. Both approaches accurately predict the Swift hardening parameters, while substantially lower predictive accuracy is obtained for the CPB'06. The FNN improves the prediction of some CPB'06 parameters compared with XGBoost, although the overall identification difficulties remain. A physics-informed loss function is also introduced to penalize predictions resulting in non-admissable lankford coefficients. This approach improves the physical admissibility of the CPB'06 parameters. The results suggest that parameter identifiability from the considered mechanical response remains a major challenge for machine learning-based calibration of CPB'06.*

1 INTRODUCTION

The accurate prediction of the mechanical behavior of sheet metals under complex loading states is essential for optimizing industrial stamping and forming processes. The predictive capability of these numerical simulations is bound to the fidelity of the adopted constitutive models, which in turn depends on the precise identification of the constitutive parameters [1].

Finite Element Model Updating (FEMU) and the Virtual Fields Method (VFM) are examples of classical inverse optimization strategies that are used to identify these parameters [2]. Although versatile and accurate, these iterative methods face challenges in computational cost, as material models become more sophisticated [1]. Machine Learning (ML) approaches have emerged as a good alternative to bypass these computational costs in solid mechanics [2]. These ML-based approaches, trained on numerical or experimental databases, can execute parameter predictions almost instantaneously, enabling real-time material characterization.

However, the transition from simpler models to advanced anisotropic formulations introduces significant challenges in parameter identifiability. The calibration of a model such as the CPB'06 yield criterion [3] is more complex than identifying the parameters of the classical quadratic Hill'48 model [1]. While Hill'48 uses a compact decoupled parameter set, the CPB'06 criterion introduces a higher dimensional space with up to nine orthotropic coefficients of the linear transformation tensor alongside the homogeneity exponent (a) and the strength parameter (k) [1]. This high dimensionality, combined with the non-linear structure of the criterion, leads to a strong mathematical coupling among the parameters. This makes the parameter identification process have a "multiplicity of solutions" phenomenon, where different parameter sets produce kinematically similar yield surfaces [1].

While numerous studies have demonstrated the successful application of machine learning for calibrating simpler criteria like Hill'48, previous attempts to apply machine learning to more advanced formulations, such as CPB'06, have not yet got satisfactory results [4]. This work investigates the performance of ML-based parameter identification for the CPB'06 yield criterion using synthetic data from biaxial tensile simulations. XGBoost and feedforward neural networks are compared, and the effects of physically constrained data generation and a physics-constrained loss function on constitutive parameter identification are evaluated.

2 MACHINE-LEARNING FRAMEWORK

The overall machine-learning framework adopted in this work is shown in Figure 1. A physically constrained material database is first generated from finite element simulations of the biaxial cruciform test. The resulting mechanical response is then processed using Principal Component Analysis (PCA), which reduces the dimensionality of the input data while retaining its main sources of variation. The reduced dataset is subsequently used to train and evaluate XGBoost and feedforward neural network models for constitutive parameter identification. Finally, a physics-constrained loss function is introduced to assess its effect on the physical admissibility of the predicted parameters. The following sections describe the numerical model, dataset generation, and learning approaches in detail.

2.1 Numerical model

A three-dimensional finite element model of a biaxial tensile test on a cruciform specimen was developed in ABAQUS/Standard to generate the synthetic dataset. The specimen geometry, previously designed to promote heterogeneous stress and strain fields, is shown in Figure 2a [1,

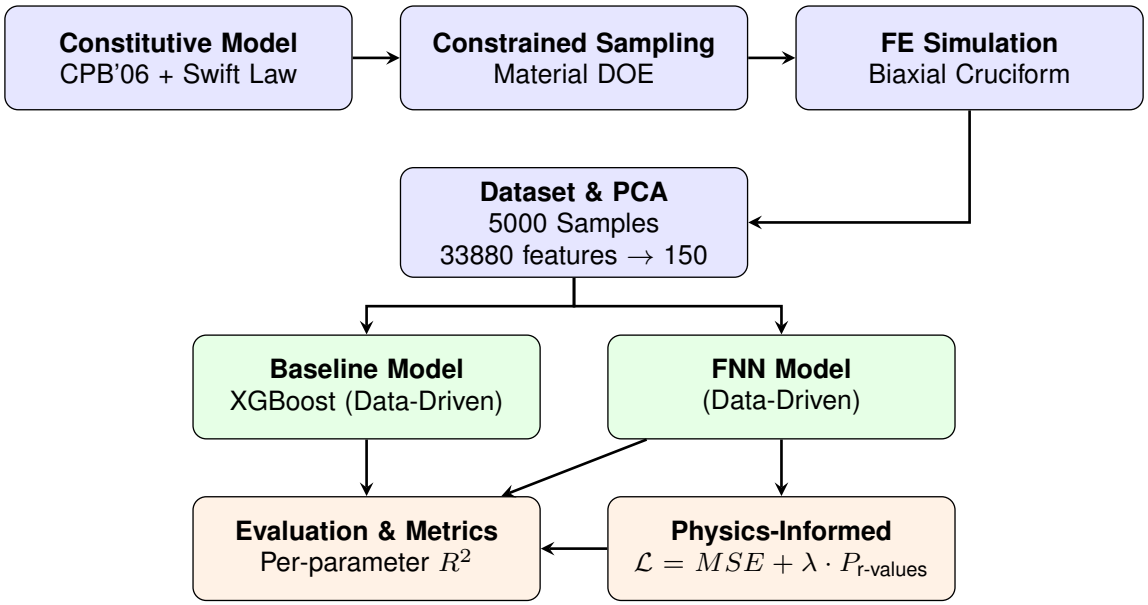


Figure 1: Overview of the machine-learning framework for CPB'06 constitutive parameter identification.

4]. To reduce the computational cost while retaining the relevant mechanical response, only one-eighth of the specimen was modeled by exploiting the material, geometric, and boundary-condition symmetries of the biaxial configuration, as shown in Figure 2b. Symmetry boundary conditions were imposed on the three orthogonal planes $x = 0$, $y = 0$, and $z = 0$, while equal prescribed displacements of $u_x = u_y = 2.0$ mm were applied to the grip regions of the two arms. The simulations were performed under static conditions and divided into 20 equally spaced increments. At each increment, the reaction forces in the xx- and yy-directions and the strain components ε_x, ε_y, and ε_{xy} were recorded. The specimen was discretized using a structured mesh with 564 C3D8R elements, based on a previous mesh-convergence study [1].

An elastoplastic constitutive model was adopted, combining isotropic linear elasticity with the anisotropic CPB'06 yield criterion and isotropic Swift hardening. Young's modulus and Poisson's ratio were set to $E = 210$ GPa and $\nu = 0.3$, respectively. The hardening response was described by the Swift law under an associated flow rule:

$$\sigma_y = K\left(\varepsilon_0 + \bar{\varepsilon}^p\right)^n, \tag{1}$$

where σ_y is the current yield stress, $\bar{\varepsilon}^p$ is the equivalent plastic strain, and K, ε_0, and n are the material hardening parameters. The initial yield stress σ_0 is given by [1]:

$$\sigma_0 = K(\varepsilon_0)^n. \tag{2}$$

The anisotropic yield behaviour was described using the CPB'06 criterion [5]:

$$B\left[(|\Sigma_1| - k\Sigma_1)^a + (|\Sigma_2| - k\Sigma_2)^a + (|\Sigma_3| - k\Sigma_3)^a\right]^{\frac{1}{a}} = \sigma_y, \tag{3}$$

where Σ_1, Σ_2, and Σ_3 are the principal values of the transformed deviatoric stress tensor $\boldsymbol{\Sigma}$, defined as $\boldsymbol{\Sigma} = \mathbf{C}\boldsymbol{\sigma}'$, where $\boldsymbol{\sigma}'$ is the deviatoric stress tensor and $\mathbf{C}$ is a is a constant fourth order tensor containing nine independent anisotropy parameters. The parameter a is a positive integer representing the degree of homogeneity of the yield function, while k is a coefficient that represents the ratio between tensile and compression yield stresses. The σ_y represents the yield stress of the material, and B is a constant that guarantees that the yield stress is identical to the tensile yield stress in the rolling direction [5].

Although the tensor $\mathbf{C}$ theoretically contains nine independent anisotropy parameters, for metallic sheets, $C_{44} = C_{55} = 1$ and $C_{11} = 1$ are assumed. In addition, the material constants $k = 0$ and $a = 2$ are selected [1]. The identification problem is reduced to six active anisotropic parameters ($C_{12}, C_{13}, C_{22}, C_{23}, C_{33}$ and C_{66}), together with the three isotropic hardening parameters (K, σ_0 and n).

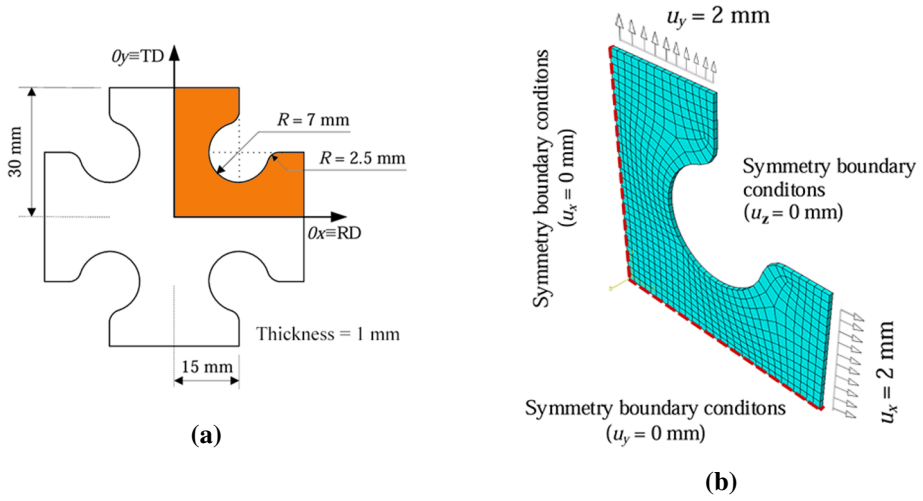


Figure 2: Overview of the biaxial tensile test setup: (a) geometry and dimensions of the cruciform sample [4], and (b) finite element mesh with applied boundary conditions.

2.2 Dataset generation

In previous studies [4], data generation relied on unconstrained sampling techniques (e.g., Latin Hypercube Sampling) applied directly to the mathematical variables of the yield criteria. However, for complex anisotropic formulations such as CPB'06, unconstrained sampling can generate physically meaningless parameter combinations, including materials exhibiting negative Lankford coefficients (r-values). In this work, a constrained data generation strategy was

implemented to ensure the physical admissibility of the generated materials.

The initial parameter space was genereted using a Sobol sequence [6] to obtain a well-distributed sampling of the prescribed constitutive parameter space, maintaining the predefined constitutive limits detailed in Table 1. An automated filtering routine was then applied to enforce the physical admissibility constraints. To prevent premature localized necking, the hardening parameters were restricted using the Considère criterion [7] applied to the Swift's Law ($n - \varepsilon_0 \geq 0.2$). This threshold guarantees that the specimen endures the applied displacement without localized failure. In addition, the CPB'06 parameters were constrained to enforce realistic planar anisotropy. The simulated Lankford coefficients (r-values), calculated at 1-degree increments, were strictly bounded between 0.6 and 5.7 [8]. Retaining only the materials that simultaneously satisfied these hardening and anisotropic restrictions yielded a final dataset of exactly 5,000 unique, physically consistent material samples. This dataset was randomly partitioned into 3,500 samples for training, 750 for validation and 750 for testing.

Each sample contains 33,880 input features, corresponding to the mechanical response recorded over 20 time steps, including two global reaction forces and three local strain components at each of the 564 finite elements. The corresponding target matrix has dimensions $n_{\text{samples}} \times 9$ and contains the six active CPB'06 anisotropy parameters and the three Swift hardening parameters.

Table 1: Parameter input space and physical constraints for the dataset generation.

Category	**Parameter / Constraint**	**Input Space / Condition**
Sampled Parameters	C_{22}, C_{33}, C_{66}	$[-0.5, 2.5]$
	C_{12}, C_{13}, C_{23}	$[-1.5, 1.5]$
	σ_0 [MPa]	$[150, 1200]$
	n	$[0.20, 0.25]$
	ε_0	$[0.002, 0.030]$
Physical Filters	Lankford coefficients (r-values)	$0.6 \leq r(\theta) \leq 5.7$
	Strain at the Onset of Necking ($n - \varepsilon_0$)	≥ 0.2

2.3 Learning models

Extreme Gradient Boosting (XGBoost) is utilized as the baseline ML regression model due to its scalability and efficiency for parameter calibration [9]. The algorithm operates on an additive tree framework, sequentially constructing weak decision trees to minimize a regularized objective function [10]. Prior to model training, the input feature matrix is standardized using z-score normalization via the `StandardScaler` algorithm. PCA was then applied to reduce the 33,880 raw features to 150 principal components. This number was the optimal dimension determined through previous sensitivity studies [9]. Since standard XGBoost is a single-output algorithm, a `MultiOutputRegressor` was used to simultaneously predict the target mate-

rial parameters. Most hyperparameters are kept at their default settings, with the exception of the learning rate ($\eta = 0.02$), maximum tree depth ($\texttt{max_depth} = 4$), and the number of estimators ($\texttt{n_estimators} = 1000$), following optimized settings from previous benchmarks [4].

As an alternative to XGBoost, a Feedforward Neural Network (FNN) was implemented to learn the nonlinear mapping between the reduced mechanical response and the nine constitutive parameters of the CPB'06 yield criterion and Swift hardening law. The same PCA-based input representation was used, resulting in 150 input features. The network hyperparameters were determined through Bayesian optimization. The resulting architecture consists of an input layer with 150 neurons, two hidden layers with 64 neurons each and Leaky ReLU activation functions, and a linear output layer with 9 neurons corresponding to the target constitutive parameters.

2.4 Performance metrics

The predictive performance of the regression models is assessed on the independent testing set by comparing the predicted and true values of each material parameter using the coefficient of determination (R^2), defined as [1]:

$$R^2 = 1 - \frac{\sum_{i=1}^{S}(A_i - \hat{A}_i)^2}{\sum_{i=1}^{S}(A_i - \bar{A})^2}, \tag{4}$$

where A_i denotes the target value of a given constitutive parameter for sample i, $\hat{A}_i$ is the corresponding predicted value, $\bar{A}$ is the mean target value over the testing set, and S is the total number of testing samples.

3 RESULTS AND DISCUSSION

3.1 Baseline performance

To establish a baseline for the machine learning-based parameter identification, an XGBoost regressor was evaluated using the CPB'06 yield criterion combined with the Swift hardening law. The XGBoost model achieved a global coefficient of determination of $R^2 \approx 0.81$ on the training dataset, wich decreased to $R^2 \approx 0.37$ on the test set. This indicates substantional overfitting and limited generalization to unseen data.

Figure 3 shows parameter-wise scatter plots comparing the true values against the XGBoost predictions for the train, validation and test sets. The isotropic hardening parameters σ_0, K and n are predicted with high accuracy, with R^2 values of 0.99, 0.99 and 0.90, respectively. This indicates that the response of the cruciform specimen contains sufficient information to identify the swift hardening parameters. In contrast, parameters C_{12}, C_{13}, C_{22}, C_{23}, C_{33} are predicted with low accuracy. This poor predictive performance may be associated with the limited sensitivity of the mechanical response to these parameters and their strong coupling within the CPB'06 formulation. Also, different parameter combinations can produce similar mechanical responses, resulting in a many-to-one mapping and poorly constrained predictions. The parameter C_{66} ($R^2 = 0.28$) shows better identifiability, although its predictive accuracy

remains limited. This may be attributed to the predominantly normal strain states induced by the orthogonal loading of the cruciform specimen, which provide limited sensitivity to the shear response.

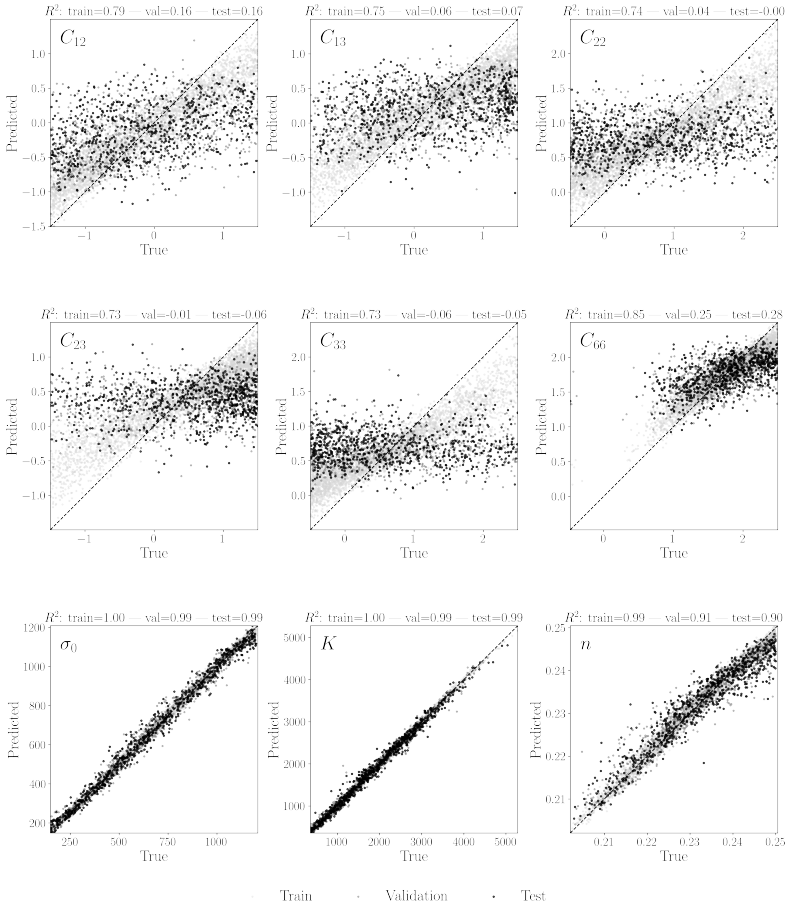


Figure 3: Scatter plots of predicted over true values of parameters of CPB'06 and Swift's Law with XGBoost.

3.2 FNN performance

To evaluate the influence of the learning algorithm on parameter identification, the XGBoost regressor was replaced with an FNN. Figure 4 presents the parameter-wise predictive performance of the FNN on the train, validation and test sets. Similarly to XGBoost model, the FNN accurately predicted the isotropic hardening parameters σ_0, K, and n with R^2 values of 0.99, 0.99 and 0.97, respectively. This further indicates response of the cruciform specimen contains sufficient information to identify the Swift hardening parameters. The FNN improves the prediction of the CPB'06 parameters compared with XGBoost, although their overall predictive accuracy remains limited. The R^2 value for C_{66} improves from 0.28 to 0.49, while C_{12} and C_{13}

reach values of 0.27 and 0.22, respectively. The remaining parameters C_{22}, C_{23}, and C_{33} show lower predictive accuracy, with R^2 values of 0.19, 0.09, and 0.04, respectively. Overall, replacing XGBoost with an FNN improves the prediction of several CPB'06 parameters but does not overcome the difficulties observed in their identification.

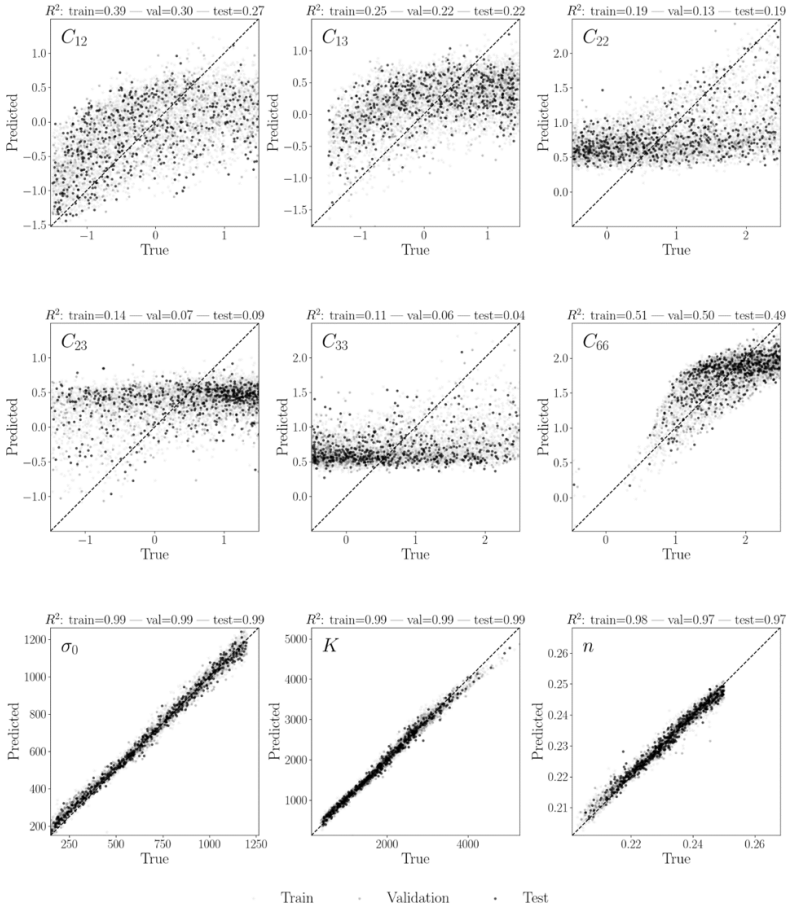


Figure 4: Scatter plots of predicted over true values of parameters of CPB'06 and Swift's Law with FNN model

3.3 Physics-Informed Learning

Although the training dataset contains only physically admissible materials, the baseline FNN predicts parameter combinations resulting in Lankford coefficients outside the prescribed range of $0.6 \leq r(\theta) \leq 5.7$, as illustrated in Figure 5(a). This highlights that physically admissible training data do not guarantee physically admissible predictions.

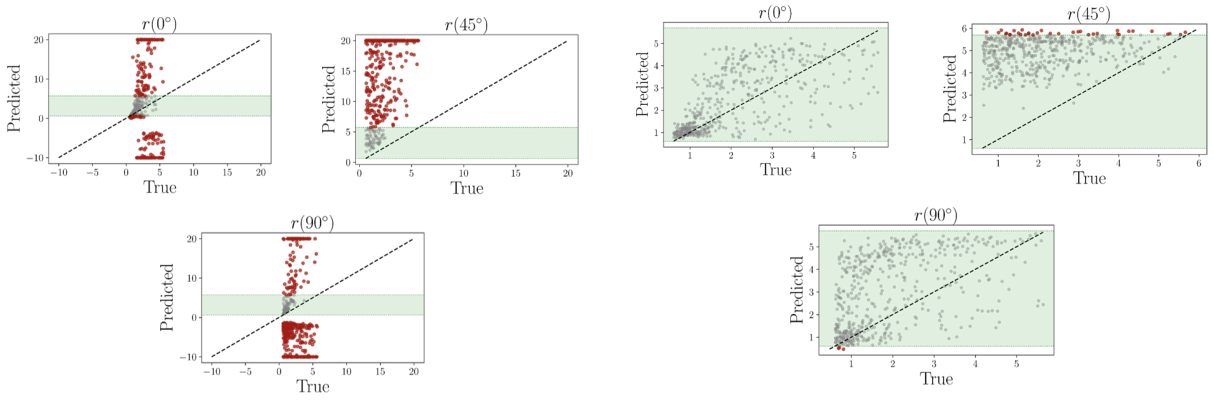


Figure 5: True and predicted Lankford values as a function of the angle with respect to the rolling direction (0°, 45°, and 90°) for the: (a) baseline model and (b) the physics-informed model with $\lambda = 2$.

To improve physical admissibility of the predicted parameters, the loss function was modified to penalize predictions outside this range:

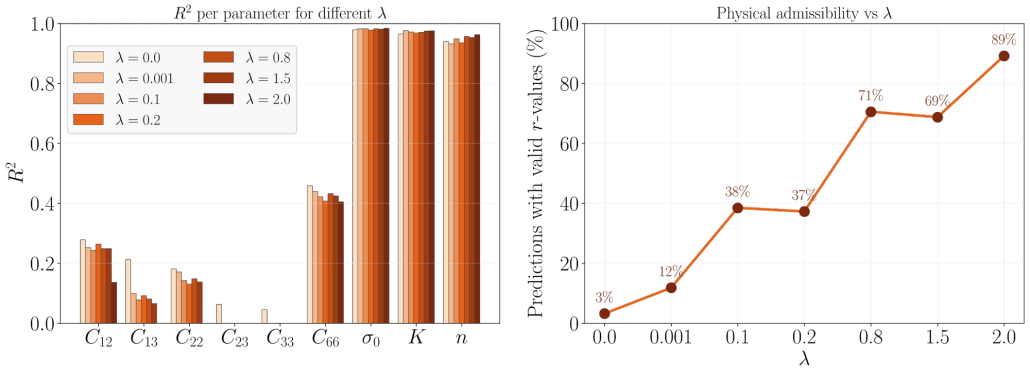
$$\mathcal{L} = \text{MSE} + \lambda \cdot P_{\text{r-values}}, \tag{5}$$

where MSE is the mean squared error between the predicted and target parameters, $P_{\text{r-values}}$ quantifies the penalty associated with predicted r-values outside the prescribed range, and λ controls the contribution of this penalty to the loss function. The penalty weight was varied as $\lambda \in \{0, 0.001, 0.1, 0.2, 0.8, 1.5, 2.0\}$, with $\lambda = 0$ represents the baseline FNN. The penalty term, $P_{\text{r-values}}$ is given by:

$$P_{r\text{-values}} = \frac{1}{N} \sum_{\theta \in \Theta} \left[\text{ReLU}(r_{\min} - r(\theta))^2 + \text{ReLU}(r(\theta) - r_{\max})^2 \right], \tag{6}$$

where $N = 7$ is the number of evaluated orientations, $\Theta = \{0°, 15°, 30°, 45°, 60°, 75°, 90°\}$, and $r_{\min}$ and $r_{\max}$ are equal to 0.6 and 5.7, respectively (see Table 1).

Increasing $\lambda = 0$ increases the percentage of predictions with physically admissible r-values (Figures 5b and 6b), However, the parameter-wise R^2 values do not improve and generally decrease with increasing $\lambda = 0$ (Figure 6a). Thus, the physics-informed loss improves the physical admissibility of the predictions but not the accuracy of the identified CPB'06 parameters.



(a) (b)

Figure 6: Effect of the value of λ: (a) the R^2 per parameter, and (b) the physical admissibility of the Lankford values predicted as a function of the value of λ.

4 CONCLUSIONS

The identification of CPB'06 parameters using machine learning remains challenging. Both XGBoost and FNN accurately predict the Swift hardening parameters, while the CPB'06 parameters show substantially lower predictive accuracy. The FNN improves the prediction of some CPB'06 parameters compared with XGBoost, but the overall identification difficulties remain.

Incorporating physical constraints into the FNN loss function increases the percentage of predictions with physically admissible r-values but does not improve the direct prediction of the CPB'06 parameters. These results suggest that the main limitation may be related to parameter identifiability from the mechanical response considered, rather than solely to the choice of machine learning algorithm. Future work will therefore explore alternative target representations and multi-test calibration strategies to improve parameter identifiability.

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REFERENCES

[1] João António Gamboa Marques and Pedro André Dias Prates. *Calibration of plasticity models for metal sheets using machine learning*. Aveiro: J. Marques, 2024. URL: https://ria.ua.pt/handle/10773/44118.

[2] A. Andrade-Campos et al. "On the inverse identification methods for forming plasticity models using full-field measurements". In: *IOP Conference Series: Materials Science and Engineering* 1238 (May 2022), p. 012059. DOI: 10.1088/1757-899X/1238/1/012059.

[3] Oana Cazacu, Brendan Plunkett, and Frédéric Barlat. "Orthotropic yield criterion for hexagonal closed packed metals". In: *International Journal of Plasticity* 22 (2006), pp. 1171–1194.

[4] Pedro Prates et al. "Evaluating the impact of 3D constitutive models on predictive performance of XGBoost for material parameter identification". In: *Materials Research Proceedings*. Vol. 54. 2025, pp. 1597–1604.

[5] Armando Marques et al. "Machine Learning Algorithms Applied to the Identification of CPB06 Yield Criterion Parameters". In: *Key Engineering Materials* 1049 (Apr. 2026), pp. 89–98. DOI: 10.4028/p-Jot8jG.

[6] Sebastian Burhenne and Dirk Jacob. "Sampling based on Sobol' sequences for Monte Carlo techniques applied to building simulations". In: Jan. 2011, pp. 1816–1823.

[7] Kerry S. Havner. "On the onset of necking in the tensile test". In: *International Journal of Plasticity* 20.4 (2004), pp. 965–978. ISSN: 0749-6419. DOI: https://doi.org/10.1016/j.ijplas.2003.05.004. URL: https://www.sciencedirect.com/science/article/pii/S0749641903001323.

[8] A. E. Marques, T. G. Parreira, A. F. G. Pereira, et al. "Machine learning application to the identification of sheet metal constitutive model parameters". In: *Machine Learning: Computer Science and Engineering* 1 (2025), p. 10. DOI: 10.1007/s44379-024-00006-8.

[9] Pedro Prates, Dário Mitreiro, and António Andrade-Campos. "Limitations of XGBoost in Predicting Material Parameters for Complex Constitutive Models". In: *MATEC Web of Conferences* 408 (May 2025), p. 01023. DOI: 10.1051/matecconf/202540801023.

[10] Tianqi Chen and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System". In: *CoRR* abs/1603.02754 (2016). arXiv: 1603.02754. URL: http://arxiv.org/abs/1603.02754.